Spatial Statistics

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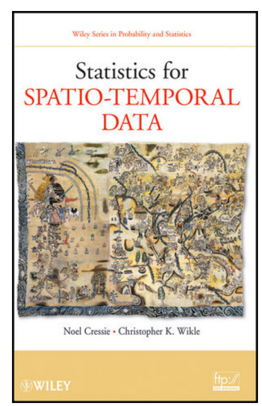
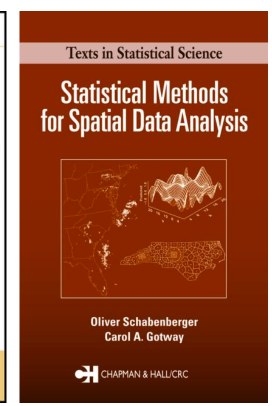
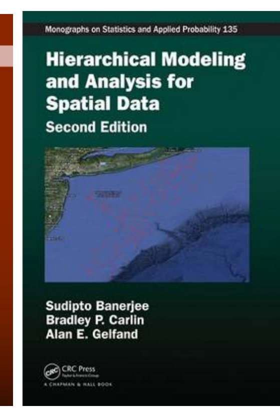
What is Spatial Statistics?

- Statistics
  - The study of ***collection***, ***analysis***, ***interpretation*** of data
  - Descriptive v.s. inferential
- Spatial statistics
  - Statistics for spatial data (**point**, line, polygon, **raster**)
  - Variables indexed in 2D or 3D, random locations
  - Unique properties
    - Non i.i.d.
    - Spatial autocorrelation
    - Isotropy v.s. anisotropy
    - Stationarity v.s. non-stationarity

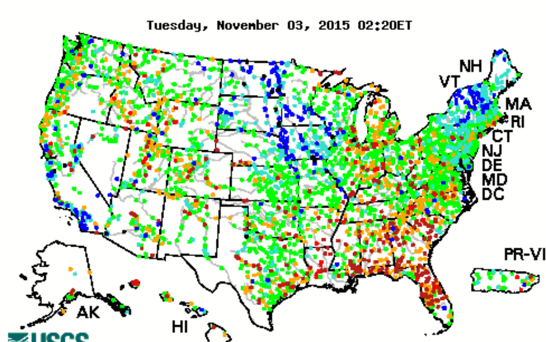




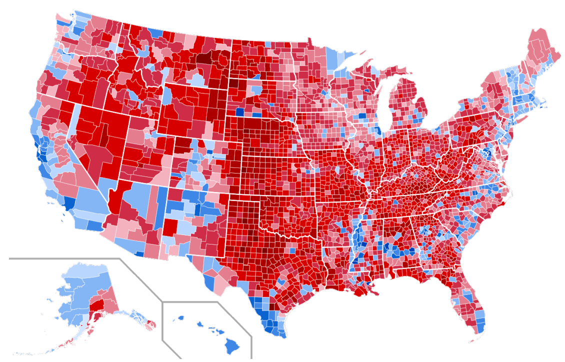


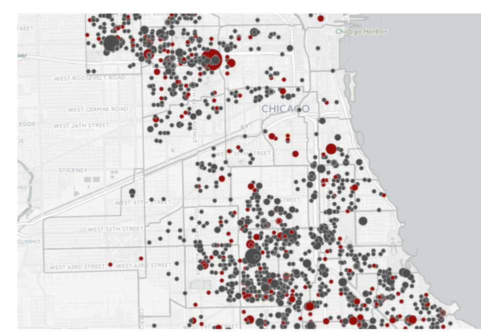
Categories of Spatial Statistics

- Geostatistics – point reference data
  - $\{Y(s): s \in D\}, D$ is r-dimensional Euclidean space
  - Y is random, s is fixed
- Lattice statistics – areal data
  - $\{Y(s): s \in D\}, D$ is a tessellation of Euclidean space
  - Y is random, s is fixed
- Spatial point process – spatial point event data
  - $\{s_1, s_2, s_3, \ldots, s_n\}, s_i'$s are point locations
  - s is random



U.S. river stream gauge observations
Source: USGS.gov

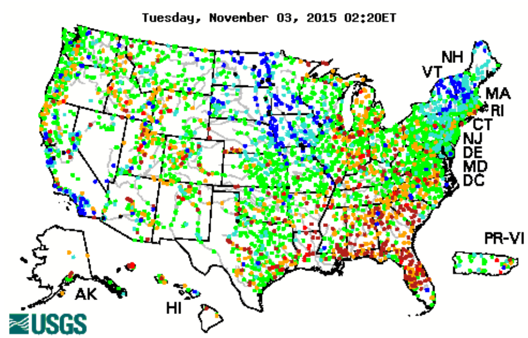

Election result by county in 2016
Source: http://brilliantmaps.com



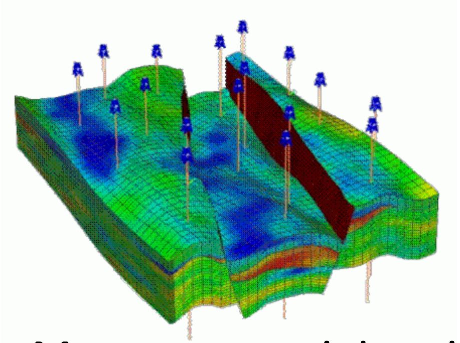
Shooting, Chicago 2010
Source: http://assets.dnainfo.com

Part I: Geostatistics

- Point reference data
  - A stochastic process: $\{Y(s): s \in D\}$, D is r-dimensional Euclidean space
  - Example:



U.S. river stream gauge observations

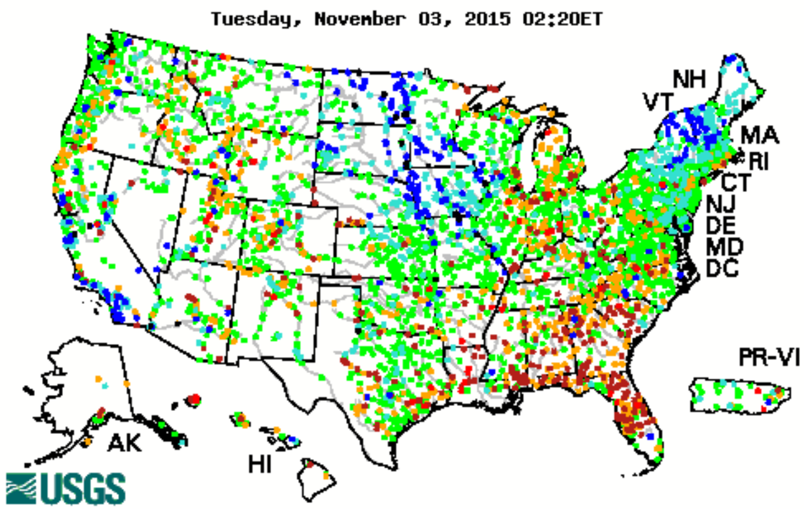


Measures at mining sites

- What is Geostatistics used for?
  - Exploratory data analysis
  - Spatial interpolation

Applications

- Estimate precipitation based on records at a set of weather stations
- Infer ground water level based on sensor readings of a set of gauges
- Predict mineral resources based on samples at a limited number of sites



U.S. river stream gauge observations

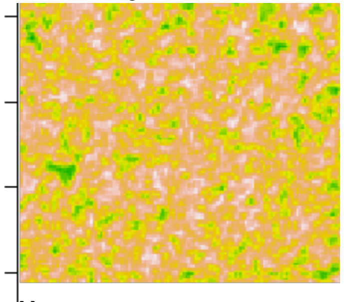
Spatial Stationarity

- $\{Y(s): s \in D\}, D$ is r-dimensional Euclidean space
- $Y(s)$ is ***strictly stationary*** when
  - *Distribution unchanged when locations shifted*
  - For any $n \geq 1$, any n locations $\{s_1, s_2, s_3, \ldots, s_n\}$, $h \in R^r$
  - $\{Y(s_1), Y(s_2), \ldots, Y(s_n)\}, \{Y(s_1+h), Y(s_2+h), \ldots, Y(s_n+h)\}$ has same distribution

Often too strong, not realistic!

- $Y(s)$ is ***weakly stationary*** when
  - *Mean, (co)variance unchanged when locations shifted*
  - $E\big(Y(s)\big) = \mu_s \equiv \mu$ (constant mean)
  - $Cov\big(Y(s), Y(s+h)\big) = C(h)$ for all $h \in R^r$

Covariance across any two locations is simply a function of h!



Illustrative example

source: http://azvoleff.com/

Variogram

- Tobler's first law of geography:
  - "Everything is related to everything else, but near things are more related than distant things."
  - How is "difference" ("irrelevance") of two observations increase with distance?
  - How to measure the range of spatial "relatedness"?
- $Y(s)$ is ***intrinsically stationary*** when
  - $E\big(Y(s)\big) = \mu_s \equiv \mu$ (constant mean)
  - $E[Y(s+h) - Y(s)]^2 \equiv 2r(h)$ for any s, h
  - $2r(h)$ is called **variogram**
  - $r(h)$ is called **semi-variogram**

"Difference" across two locations only depends on h!

- $Y(s)$ is ***isotropy*** if $r(h) \equiv r(|h|)$

With isotropy, can plot a curve for r(|h|)!

But is this assumption always valid?

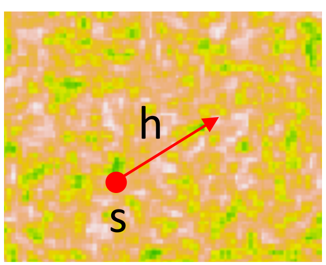
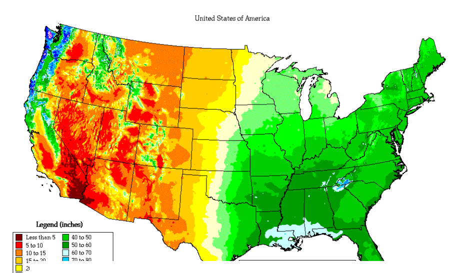


Illustration of first law of geography

source: http://azvoleff.com/



Avg. annual precip.(wrcc.dri.edu)

Variogram Plot

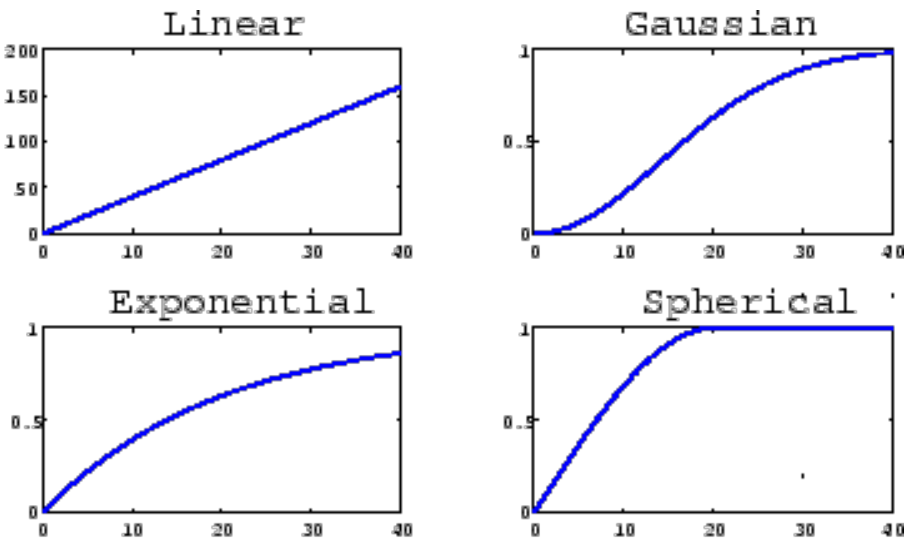
- Models of semi-variogram

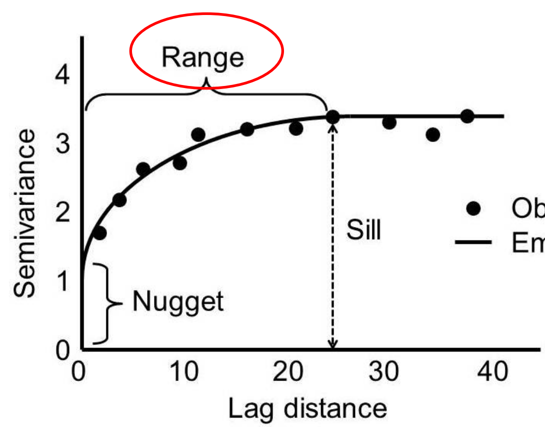
$$r(d) = \begin{cases} \tau^2 + \sigma^2 \; if \; d > 0 \\ 0 \qquad otherwise \end{cases}$$

Linear

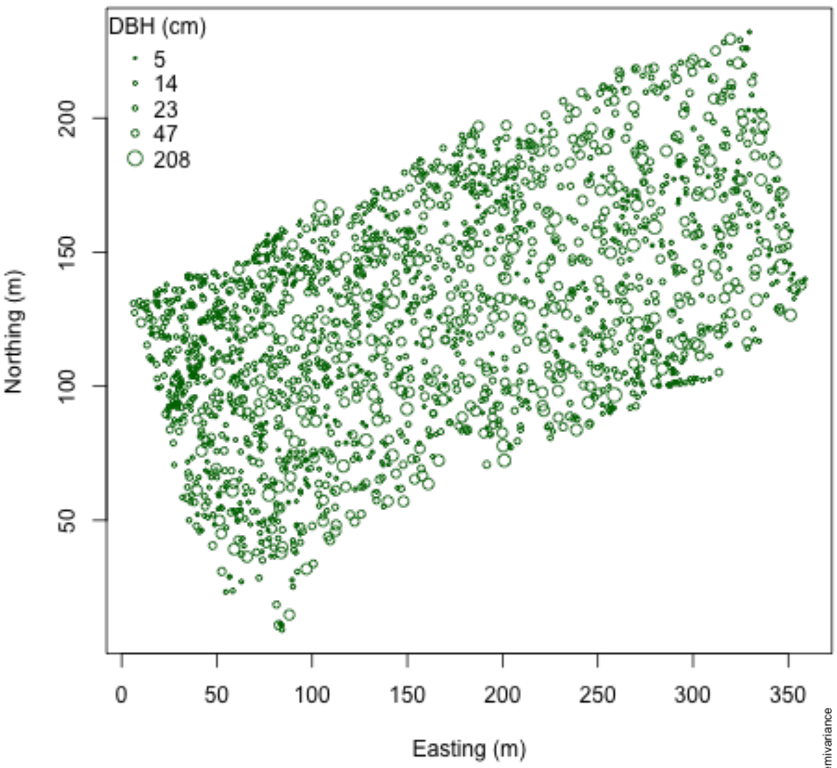
$$r(d) = \begin{cases} \tau^2 + \sigma^2 & if \; d > \frac{1}{\phi} \\ \tau^2 + \sigma^2 \frac{3\phi d - (\phi d)^2}{2} & if \; 0 < d \le \frac{1}{\phi} \\ 0 & otherwise \end{cases}$$

Spherical

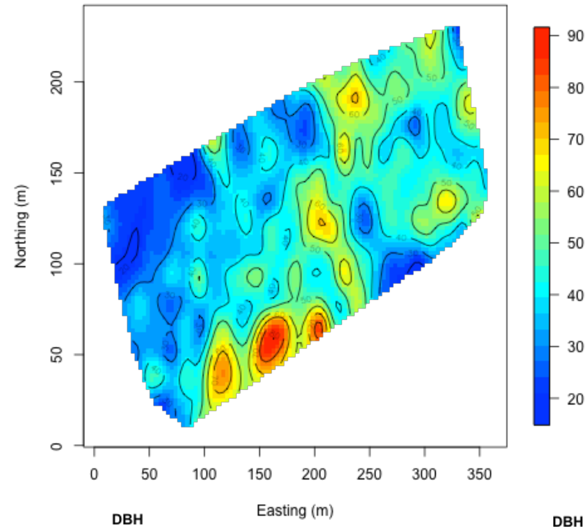


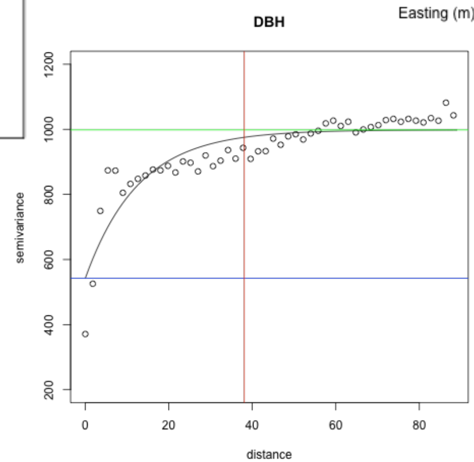


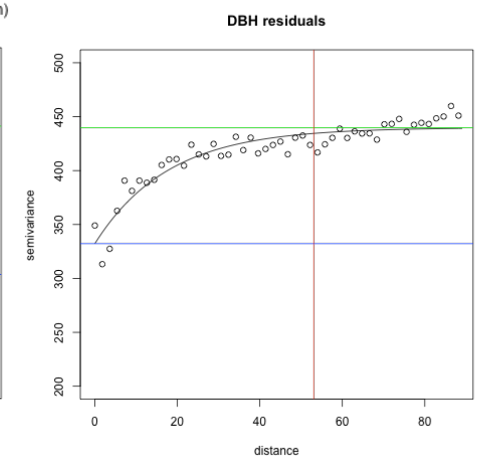
Variogram Example



Diameter at breast height on trees



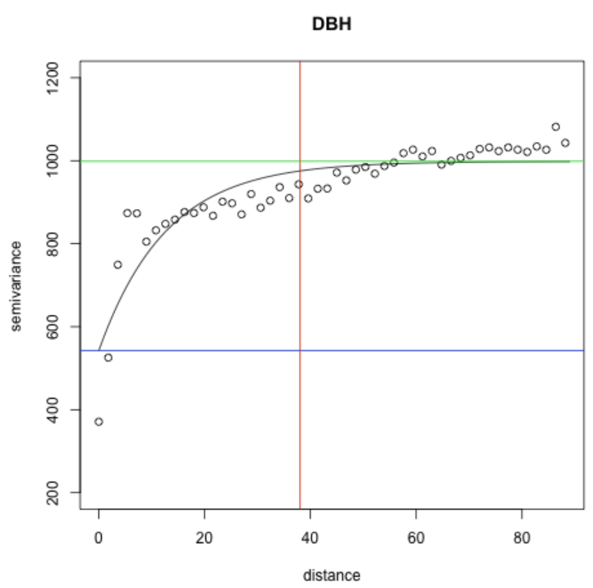




See R example!

Variogram v.s. Covarigram

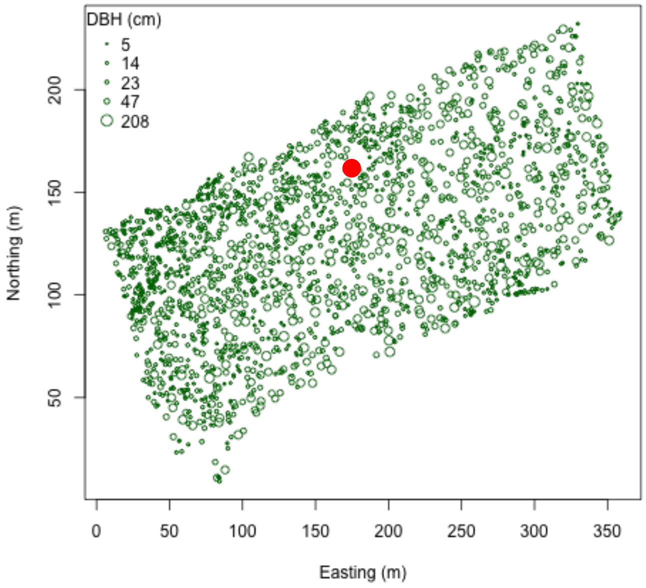
- $E[Y(s+h) - Y(s)]^2 = 2r(h)$
- $E[Y(s+h) - Y(s)]^2 = Var(s+h) + Var(s) - 2Cov(s+h, s) = 2Cov(0) - 2Cov(h)$
- Thus, $r(h) = C(0) - C(h)$, r(h) and $C(h)$ related!
- Covariance $C(h)$ is a function of h



Ordinary Kriging

- Problem:
  - Given observations at locations $\{y(s_1), y(s_2), \ldots, y(s_n)\}$,
  - How to predict $y(s_0)$
- Assumptions:
  - Weak (intrinsic) stationarity
  - ***Known covariance $\boldsymbol{C(h)}$***
  - Unknown constant $E(Y) \equiv \mu$
  - Linear estimation:
    $\hat{y}(s_0) = \sum_{i=1}^{n} l_i y(s_i)$
- Approach:
  - Minimize expected square loss!

$$E(y(s_0) - \sum_{i=1}^{n} l_i y(s_i))^2$$



Ordinary Kriging

Minimize: $E(y(s_0) - \sum_{i=1}^{n} l_i y(s_i))^2$

$$= Var(y(s_0)) - 2Cov(y(s_0), \sum_{i=1}^{n} l_i y(s_i)) + \sum_{i=1}^{n} \sum_{j=1}^{n} l_i l_j Cov(y(s_i), y(s_j))$$

$$= C_{0,0} - 2\sum_{i=1}^{n} l_i C_{0,i} + \sum_{i=1}^{n} \sum_{j=1}^{n} l_i l_j C_{i,j}$$

Notation: $C_{i,j} \equiv Cov(Y(s_i), Y(s_j))$

$$= C_{0,0} - 2C_{0,*}^T l + l^T C l$$

Remember $Cov(Y(s), Y(s+h)) \equiv C(h)$
$C(h)$ covariogram can be estimated from data!

$$\left.\begin{array}{l} E(y(s_0) - \sum_{i=1}^{n} l_i y(s_i)) = 0 \\ E(y(s_0)) \equiv E(y(s_i)) \equiv \mu \end{array}\right] \Longrightarrow 1 - \sum_{i=1}^{n} l_i = 0 \Longrightarrow \mathbf{1}^T l - 1 = 0$$

Constrained convex optimization problem! Using Lagrangian multiplier,
Optimal solution:

$$l^* = C^{-1}[C_{0*} - \frac{\mathbf{1}^T C^{-1} C_{0*} - 1}{\mathbf{1}^T C^{-1} \mathbf{1}} \mathbf{1}] \qquad \hat{y}(s_0) = l^{*T} Y$$

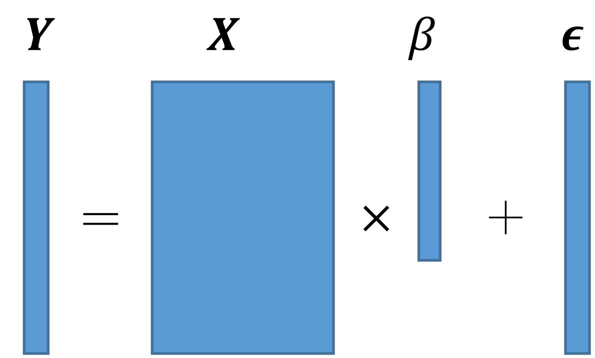
Universal Kriging

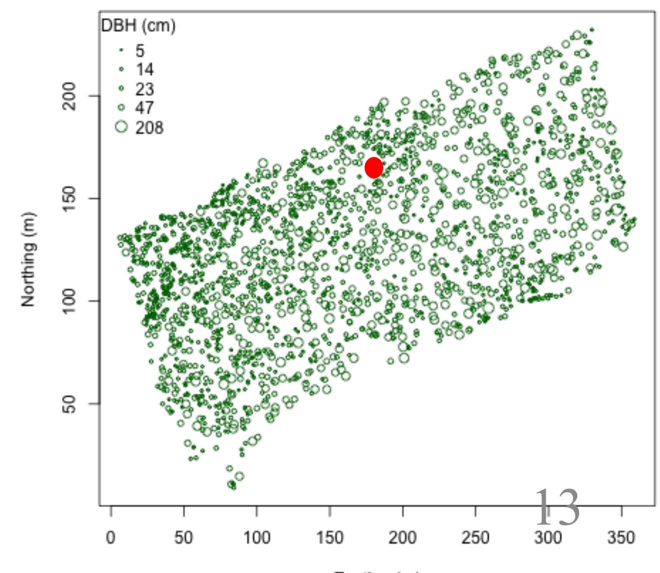
- Problem:
  - Given observations y $= \{y(s_1), \dots, y(s_n)\}$, covariates $\{x(s_1), \dots, x(s_n)\}$, $x(s_0)$, predict $y(s_0)$
- Assumptions:
  - $y(s_i) = x(s_i)^T\beta + \epsilon_i$, $\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{\epsilon}$
  - $\boldsymbol{\epsilon} \sim N(0, \Sigma)$, where $\Sigma = \sigma^2 H(\emptyset) + \tau^2 I$
- Estimator: $\hat{y}(s_0) = h(y)$
- How to find optimal $h(y)$?
  - Minimize expected square loss!

$$E(y(s_0) - h(y))^2$$

$\Longrightarrow$ Optimal prediction:

$$\hat{y}(s_0) = h(y) = E(y(s_0)|y)$$

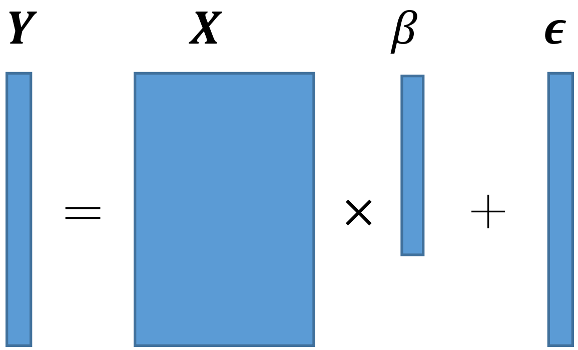




Universal Kriging

Assuming a Gaussian process, i.e., any set of observation Y follows Gaussian distribution

$\boldsymbol{Y} = \boldsymbol{X}\beta + \boldsymbol{\epsilon} \quad \boldsymbol{\epsilon} \sim N(0, \Sigma)$, where $\Sigma = \sigma^2 H(\emptyset) + \tau^2 I$



$$\begin{pmatrix} Y_1 \\ Y_2 \end{pmatrix} \sim N\left(\begin{pmatrix} \mu_1 \\ \mu_2 \end{pmatrix}, \begin{pmatrix} \Omega_{11} & \Omega_{12} \\ \Omega_{21} & \Omega_{22} \end{pmatrix} \right)$$

$$\Longrightarrow \quad E(Y_1 | Y_2) = \mu_1 + \Omega_{12}\Omega_{22}^{-1}(Y_2 - \mu_2)$$

$$Var(Y_1 | Y_2) = \Omega_{11} - \Omega_{12}\Omega_{22}^{-1}\Omega_{21}$$

Optimal prediction: $\hat{y}(s_0) = h(y) = E(y(s_0)|y)$

$$Y_1 = y(s_0)$$
$$Y_2 = y = (y(s_1), \ldots, y(s_n))^T$$
$$\mu_1 = x(s_0)^T\beta$$
$$\mu_2 = (x(s_1)^T\beta, \ldots, x(s_n)^T\beta)^T = \boldsymbol{X}\beta$$

$$\Longrightarrow \quad E(y(s_0)|y) = x(s_0)^T\beta + C_{0*}^T\Sigma^{-1}(y - X\beta)$$

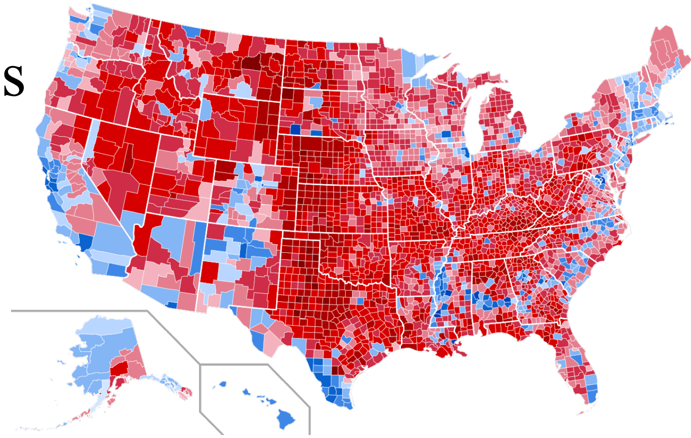
$\hat{\beta}$ $\qquad C(h) + \tau^2 I$

Review Questions:

- For each of the following statement, True or False?
  1. In Geostatistics, strict stationarity is often assumed.
  2. Variogram can help select a distance threshold for spatial neighborhood (range of spatial dependency).
  3. Kriging assumes weak or intrinsic stationarity.
  4. Ordinary Kriging assumes weak or intrinsic stationarity.

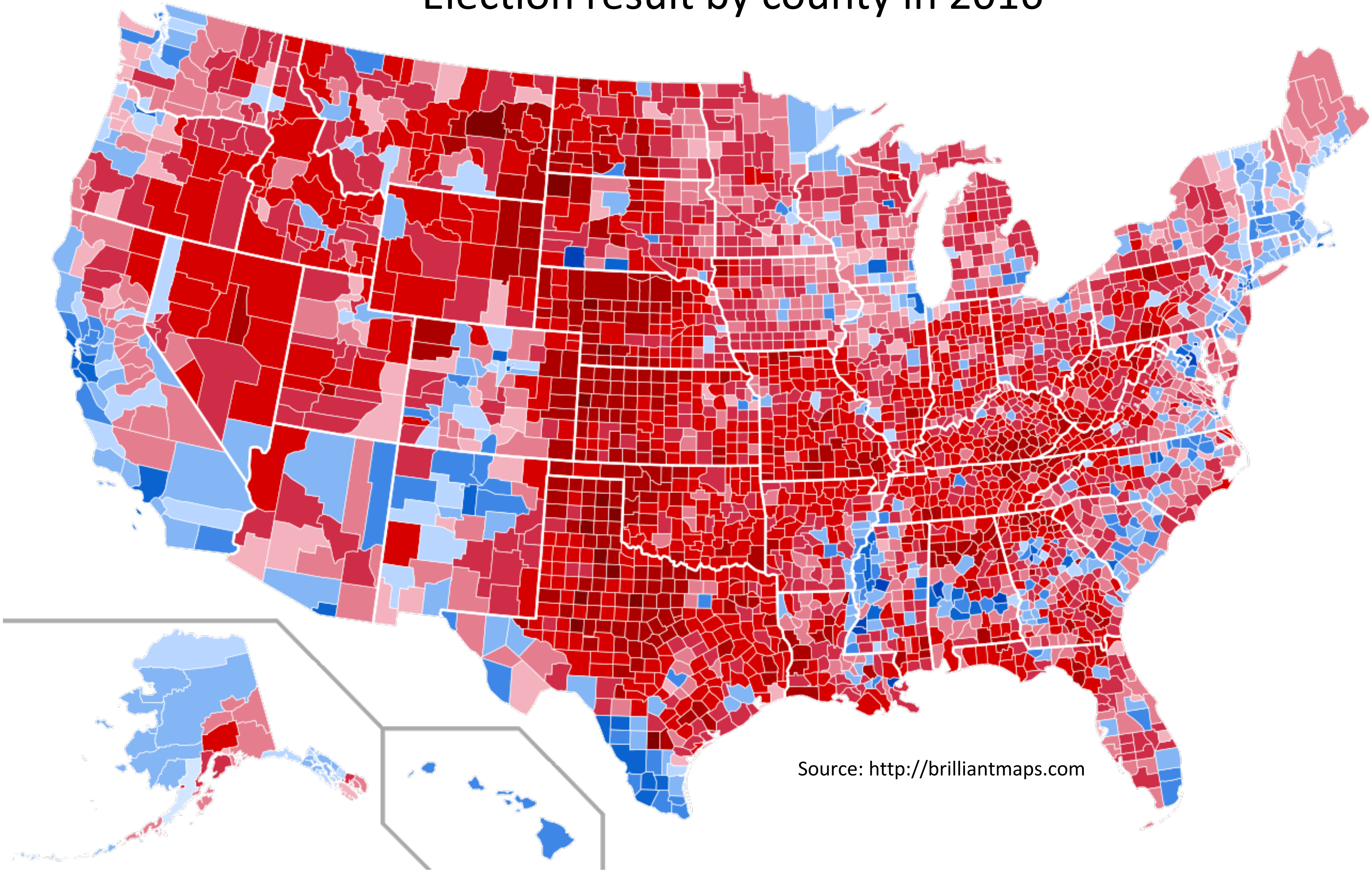
Part II: Lattice Statistics

- Areal data model
  - A tessellation of continuous space into (regular or irregular) cells
  - Mapping each unit to a non-spatial attribute value
- Lattice Statistics:
  - $\{Y(s): s \in D\}, D$ is a set of cells in areal data
- What is Lattice Statistics used for?
  - Explore spatial patterns in areal maps
  - Model areal maps for interpretation



http://brilliantmaps.com

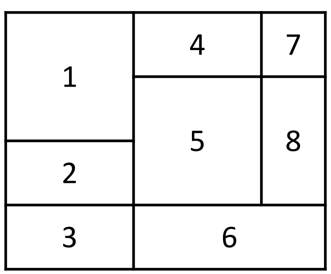
Election result by county in 2016



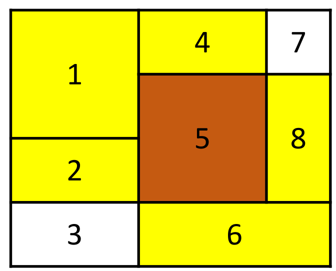
Q1: Is the map spatially stationary?
Q2: Is there strong autocorrelation globally and locally?
Q3: How to model and interpret coefficients that impact results?

W-Matrix

- Spatial neighborhood matrix
  - Wij > 0 when i and j are neighbors
  - Wij = 0 when i and j are not neighbors
- Example

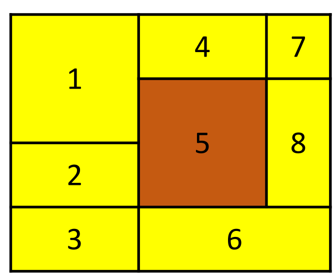


An areal data with 8 units



A rook neighborhood

0	1	0	1	1	0	0	0
1	0	1	0	1	0	0	0
0	1	0	0	0	1	0	0
1	0	0	0	1	0	1	0
1	1	0	1	0	0	0	1
0	0	1	0	1	0	0	1
0	0	0	1	0	0	0	1
0	0	0	0	1	1	1	0



A queen neighborhood

0	1	0	1	1	0	0	0
1	0	1	0	1	1	0	0
0	1	0	0	1	1	0	0
1	0	0	0	1	0	1	1
1	1	1	1	0	1	1	1
0	1	1	0	1	0	0	1
0	0	0	1	1	0	0	1
0	0	0	1	1	0	1	0

Spatial Autocorrelation

- Measures the level of global spatial association
- Moran's I:
  - $I = \frac{n \sum_i \sum_j w_{ij}(Y_i - \bar{Y})(Y_j - \bar{Y})}{\left(\sum_{i \neq j} w_{ij}\right) \sum_i (Y_i - \bar{Y})^2}$ where i and j are locations.
  - $I \in [-1, 1]$, high value shows strong spatial association
  - Example with rook neighborhood:

0	1	0	1	0
1	0	1	0	1
0	1	0	1	0
1	0	1	0	1
0	1	0	1	0

$I \approx -1$

1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
1	1	1	0	0

$I \approx 1$

Spatial Autocorrelation

- Geary's C
  - $C = \frac{(n-1)\sum_i \sum_j w_{ij}(Y_i - Y_j)^2}{2(\sum_{i \neq j} w_{ij}) \sum_i (Y_i - \bar{Y})^2}$ where i and i are locations.
  - $C \geq 0$, low values show strong spatial association
  - Example with rook neighborhood:

0	1	0	1	0
1	0	1	0	1
0	1	0	1	0
1	0	1	0	1
0	1	0	1	0

$C = ?$

1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
1	1	1	0	0
1	1	1	0	0

$C = ?$

Which one has higher C?

Local Spatial Autocorrelation

- Local indicator of spatial association (LISA)
- When data is not homogeneous, local behaviors may differ from global behavior (outliers)
- Local Moran's I:
  - $I_i = \frac{Y_i - \bar{Y}}{m_2} \sum_j w_{ij}(Y_j - \bar{Y})$ where $m_2 = \frac{\sum_i (Y_i - \bar{Y})^2}{n}$
  - $I = \frac{1}{n} \sum_i I_i$
- Local Geary's C
  - $C_i = \frac{1}{m_2} \sum_j w_{ij}(Y_i - Y_j)^2$
  - $C \propto \sum_i C_i$

Spatial Autocorrelation for Nominal Data

- Black-Black Joint Count

W	B	W	B	W
B	W	B	W	B
W	B	W	B	W
B	W	B	W	B
W	B	W	B	W

B	B	B	W	W
B	B	B	W	W
B	B	B	W	W
B	B	B	W	W
B	B	B	W	W

Suppose n locations, n_W white, n_B black.

$$P_B = \frac{n_B}{n}, P_W = \frac{n_W}{n}$$

$$JC_{BB} = \frac{1}{2}\sum_i \sum_j w_{ij} I(y_i = B, y_j = B),$$

H_0: B and W are ***independently distributed***, JC_{BB} asymptotically normal distribution

H_1: B and W are not independent, BB tends to cluster.

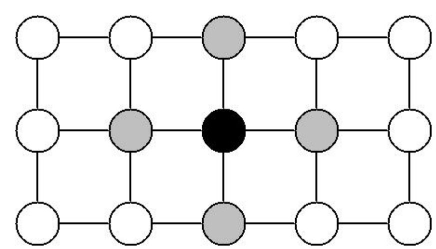
Test: $\frac{JC_{BB} - E(JC_{BB})}{\sigma}$

$E(JC_{BB}) = \frac{1}{2}\sum_i \sum_j w_{ij} P_B P_B$, $Var(JC_{BB}) = \sigma^2$ assuming Gaussian distribution

$E(JC_{BB})$ and $Var(JC_{BB})$ can also be generated from random permutation!

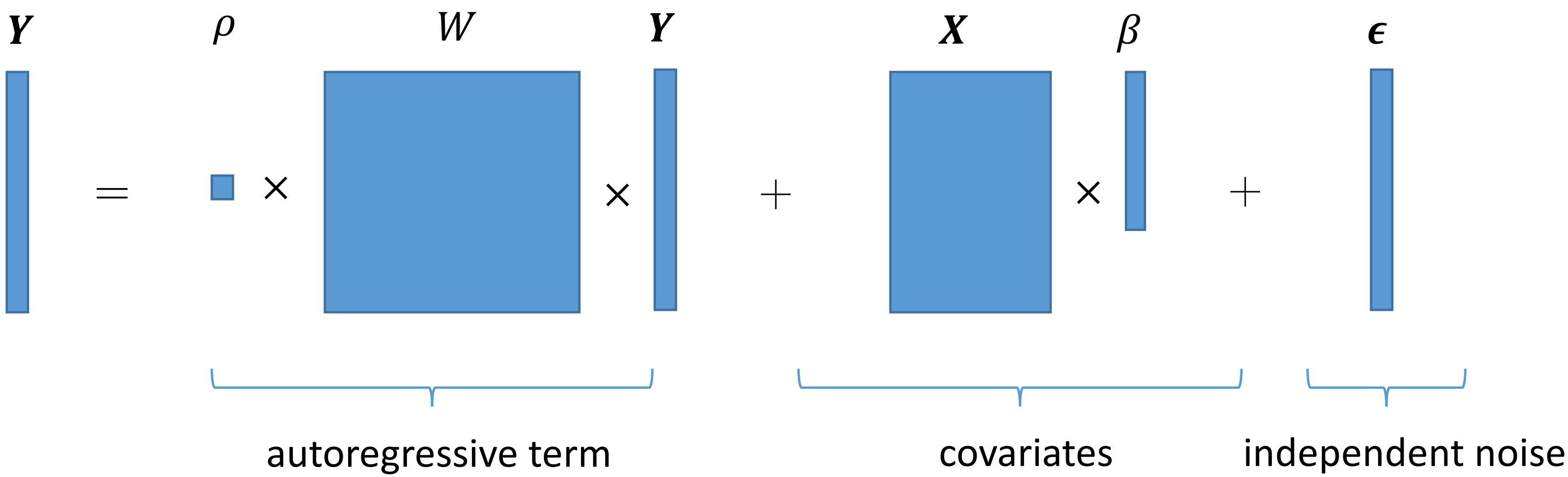
Markov Random Field

- Problem: How to model joint distribution of a field?
- Brook's Lemma:
  - Joint distribution can be determined by conditional distribution
  - $p(y_1, y_2, \ldots, y_n) \Leftarrow p\left(y_i | y_j, j \neq i\right)$
  - However, conditional distribution above is too complex!
- Markov property:
  - $p\left(y_i | y_j, j \neq i\right) \equiv p\left(y_i | y_j, j \in N(i)\right)$
  - Conditional distribution of observation at a location only depends on its ***neighbors***



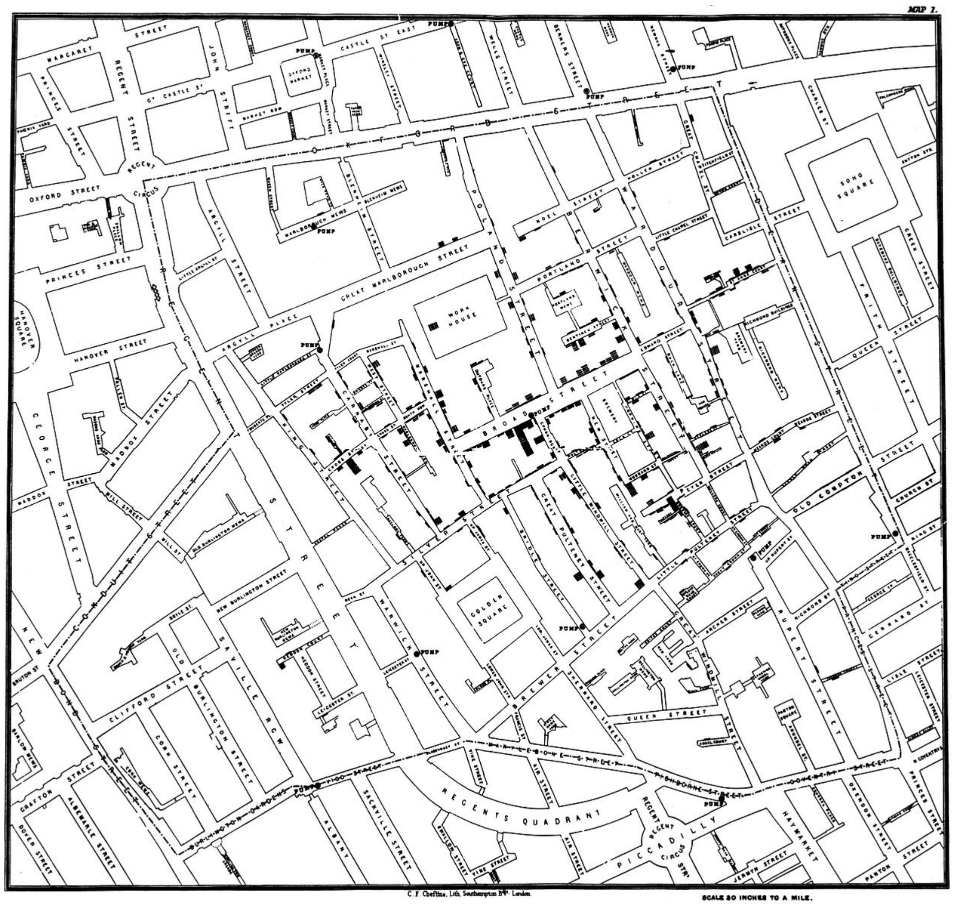
Spatial Autoregressive Model (SAR)

- $Y = \rho W Y + X\beta + \epsilon$



Part III: Spatial Point Process

- Spatial point process – spatial point event data
  - $\{s_1, s_2, s_3, \dots, s_n\}$, s_i's are event locations, fixed event type

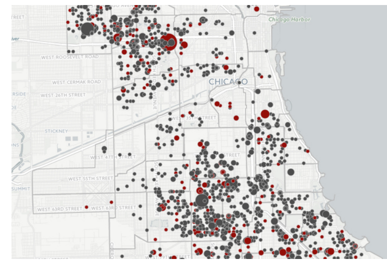


1854 Broad Street cholera outbreak
(Solid black rectangles show victims)

Part III: Spatial Point Process

Spatial Point Process

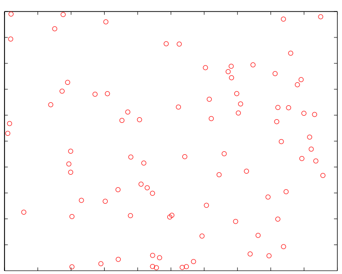
- Example
  - Crime event locations
  - Disease event locations



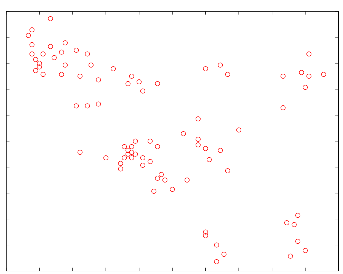
Shooting, Chicago 2010

Source: http://assets.dnainfo.com

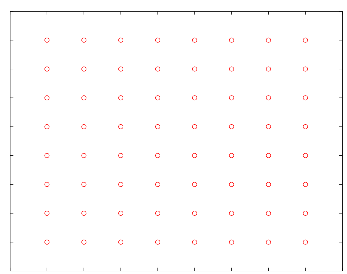
- Questions to answer
  - Are points tend to cluster/de-cluster? K-function statistics
  - Is point intensity homogeneous or there is a hotspot? Spatial scan statistics



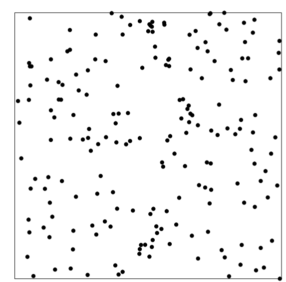
complete spatial random (CSR)



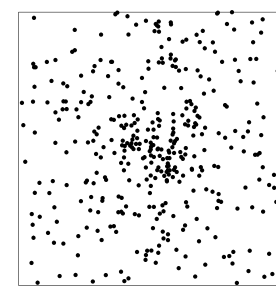
clustering



declustering



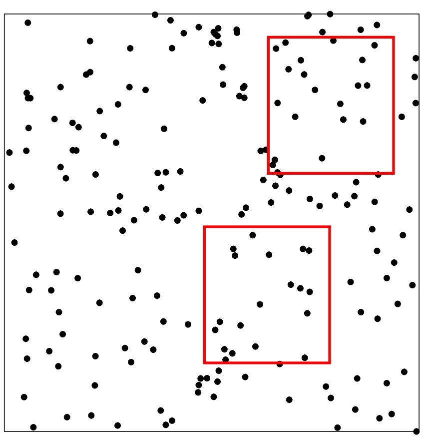
CSR



Hotspot

Homogeneous Poisson Point Process

- Complete spatial randomness (CSR)
- Intensity parameter λ, $N(A)$ as number of points in an area A
  - For any A, $N(A)$ follows $\text{Poisson}(\lambda A)$
  - For any disjoint A_1 and A_2, $N(A_1)$ and $N(A_2)$ are independent
  - In any area A, conditioned on $N(A) = n$, these n points independent and uniformly distributed in A

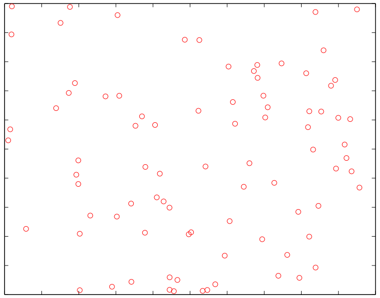


What can we say about CSR?

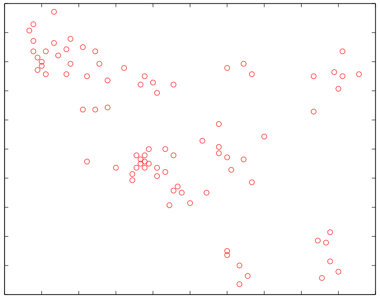
- The intensity of points are the same everywhere
- Once $N(A)$ is realized, point locations are independent
- CSR is often used as a ***null distribution***

Ripley's K Function

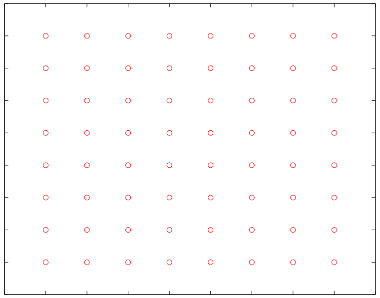
- Test if points tend to cluster with each other (micro)
- Hypothesis testing
  - H0: homogeneous Poisson point process (independent)
  - H1: points tend to cluster with each other
  - Test statistic:
    - $K(d) = \lambda^{-1} E(\#\ of\ points\ within\ radius\ d\ of\ a\ point)$
    - $\widehat{K}(d) = \lambda^{-1} \sum_{i \neq j} I(d_{ij} \leq d)/n$
    - Under H0, $K(d) = \pi d^2$



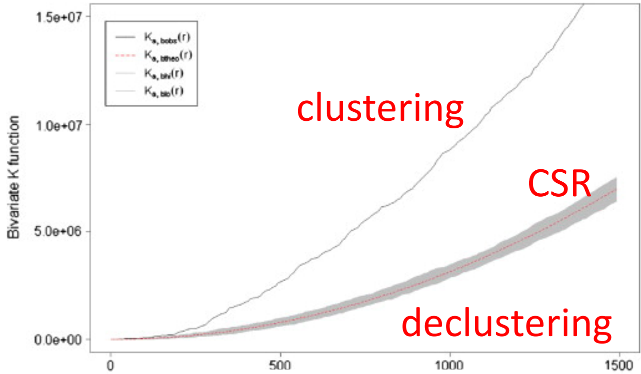
CSR



Clustering



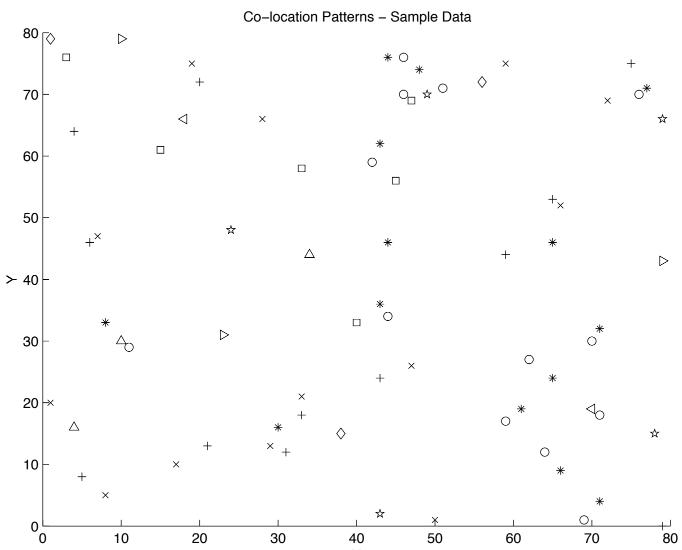
Declustering

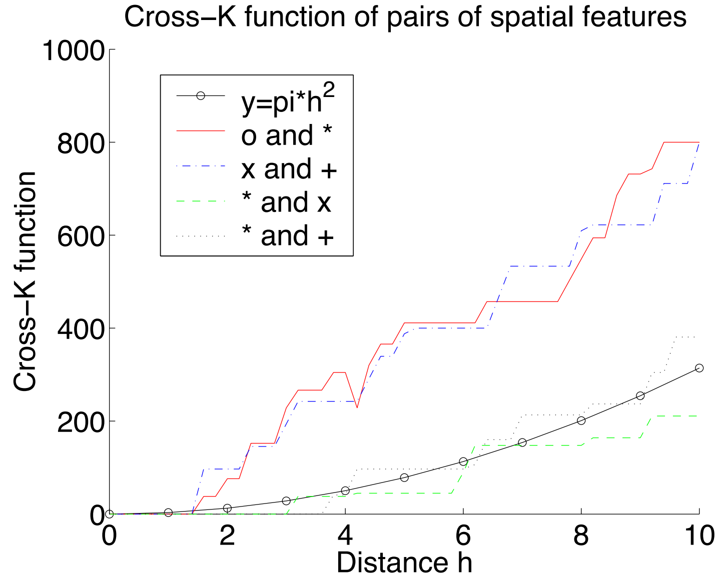


Example of K function plot

Ripley's Cross K Function

- Test if of two types of events tend to cluster together
  - H_0: event types i and j are independent
  - H_1: event types i and j tend to cluster together
  - Test statistic:
    - $K_{ij}(d) = \lambda_j{}^{-1} E(\#\, of\, points\, of\, type\, j\, within\, d\, of\, a\, point\, i)$
    - $\widehat{K_{ij}}(d) = \lambda_j{}^{-1} \sum_{i \neq j} I(d_{ij} \leq d)/n_i = (\lambda_i \lambda_j A)^{-1} \sum_{i \neq j} I(d_{ij} \leq d)$
    - Under H_0, $K_{ij}(d) = \pi d^2$





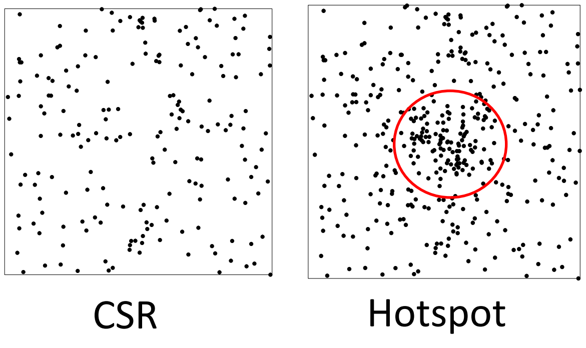
Spatial Scan Statistics

- Test if point intensity is homogeneous everywhere
- Hypothesis testing (assume Poisson point process)
  - H_0: homogeneous intensity $\lambda_{in} = \lambda_{out}$ for window W
  - H_1: inhomogeneous intensity $\lambda_{in} > \lambda_{out}$ for window W
  - Test statistic (likelihood ratio)
    - $LR = \frac{MaxLikelihood(Data|H_1)}{MaxLikelihood(Data|H_0)} = \frac{\mathrm{Sup}_{W,\lambda_{in}>\lambda_{out}} \mathrm{L}(Data; W, \lambda_{in}, \lambda_{out})}{\mathrm{Sup}_{\lambda_{in}=\lambda_{out}} \mathrm{L}(Data; W, \lambda_{in}, \lambda_{out})}$
    - $LR = \mathrm{Sup}_W \left(\frac{n_W}{E_W}\right)^{n_W} \left(\frac{n'_W}{E'_W}\right)^{n'_W} I(.)$

n_W: observed # in W n'_W: observed # out of W

E_W: expected # in W E'_W: expected # out of W

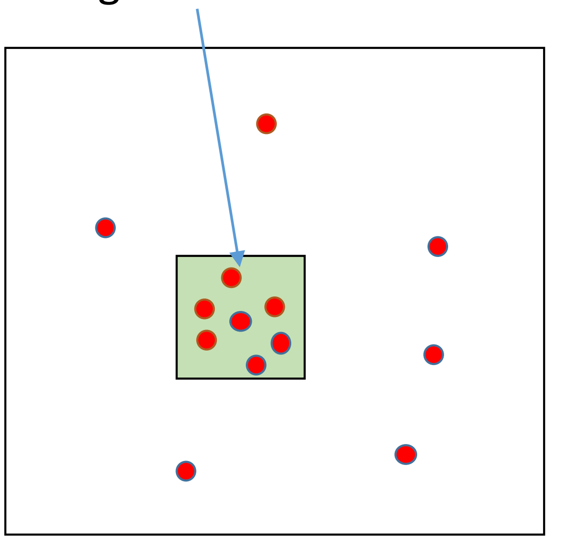
  - Significance
    - P-value, Monte Carlo simulation



Spatial Scan Statistics: An Example

- Input: 13 points
- Compute test statistic:
  - $LR = \text{Sup}_W \left(\frac{n_W}{E_W}\right)^{n_W} \left(\frac{n'_W}{E'_W}\right)^{n'_W} I(.)$
  - $n_W = 7$ (observed # in W)
  - $n'_W = 6$ (observed # out of W)
  - $E_W = \frac{13}{16} * 1$ (expected # in W)
  - $E'_W = \frac{13}{16} * 15$ (expected # out of W)
  - $I(.) = 1$ (density in W is higher than out)
- $LR = (\frac{7}{13/16})^7 (\frac{6}{13*15/16})^6 = 56115.15$
- Monte Carlo simulation

Enumerating window W with size 1



Study area size 4*4=16